Coordinate Geometry

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Distance and midpoint formulas**

Find the midpoint of the segment connecting $(-4, 7)$ and $(6, -3)$.

Midpoint $= \left(\frac{-4+6}{2}, \frac{7+(-3)}{2}\right) = (1, 2)$.

2 **[Foundational]** • **Target: Slope and equations of lines**

Write the equation of the line through $(2, -3)$ with slope $\frac{1}{2}$, in slope-intercept form.

$y - (-3) = \frac{1}{2}(x - 2)$, so $y + 3 = \frac{1}{2}x - 1$, giving $y = \frac{1}{2}x - 4$.

3 **[Foundational]** • **Target: Parallel and perpendicular lines**

Find the equation of the line parallel to $y = 4x - 1$ passing through $(0, 6)$.

Parallel lines share the same slope: $m = 4$. Since $(0, 6)$ is the y-intercept, the equation is $y = 4x + 6$.

4 **[Foundational]** • **Target: Parallel and perpendicular lines**

Find the equation of the line perpendicular to $y = -\frac{2}{3}x + 5$ passing through $(4, 1)$.

The perpendicular slope is the negative reciprocal: $m = \frac{3}{2}$. $y - 1 = \frac{3}{2}(x - 4)$, so $y = \frac{3}{2}x - 6 + 1 = \frac{3}{2}x - 5$.

5 **[Foundational]** • **Target: The equation of a circle**

Write the equation of a circle with center $(3, -2)$ and radius 7.

$(x - 3)^2 + (y + 2)^2 = 49$.

6 **[Foundational]** • **Target: The equation of a circle**

Find the center and radius of the circle $x^2 + y^2 - 6x + 4y - 3 = 0$, by completing the square.

Group and complete the square: $(x^2 - 6x + 9) + (y^2 + 4y + 4) = 3 + 9 + 4$, giving $(x - 3)^2 + (y + 2)^2 = 16$. Center: $(3, -2)$. Radius: $\sqrt{16} = 4$.

7 **[Foundational]** • **Target: Reflections, translations, and symmetry**

Find the coordinates of the point $(5, -3)$ after a reflection over the x-axis, then after a reflection over the y-axis.

Reflection over the x-axis: $(5, 3)$ (the y-coordinate's sign flips). Reflection over the y-axis (applied to the original point): $(-5, -3)$ (the x-coordinate's sign flips).

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Reflections, translations, and symmetry**

A point (a, b) is translated 3 units right and 5 units down. Write the coordinates of the image point.

$(a + 3, b - 5)$.

9 **[Intermediate]** • **Target: Finding area using the coordinate plane**

Find the area of the triangle with vertices $(0, 0)$, $(6, 0)$, and $(0, 4)$.

Since the triangle has a right angle at the origin, with legs along the axes: Area $= \frac{1}{2}(6)(4) = 12$.

10 **[Intermediate]** • **Target: Finding area using the coordinate plane**

A rectangle has vertices $(1, 1)$, $(1, 5)$, $(7, 5)$, and $(7, 1)$. Find its area and perimeter.

Width: $7 - 1 = 6$. Height: $5 - 1 = 4$. Area $= 6 \times 4 = 24$. Perimeter $= 2(6 + 4) = 20$.

11 **[Intermediate]** • **Target: Determining if points are collinear**

Determine whether the points $(1, 2)$, $(3, 6)$, and $(5, 10)$ are collinear, by comparing slopes.

Slope from $(1, 2)$ to $(3, 6)$: $\frac{6-2}{3-1} = \frac{4}{2} = 2$. Slope from $(3, 6)$ to $(5, 10)$: $\frac{10-6}{5-3} = \frac{4}{2} = 2$. Since both slopes are equal, the three points are collinear.

12 **[Intermediate]** • **Target: Finding the intersection of two lines**

Find the point of intersection of $y = 2x - 3$ and $y = -x + 9$.

Set equal: $2x - 3 = -x + 9$, so $3x = 12$, giving $x = 4$. Then $y = 2(4) - 3 = 5$. Intersection: $(4, 5)$.

13 **[Intermediate]** • **Target: Combining coordinate geometry concepts**

This question has two parts. Triangle ABC has vertices $A(0, 0)$, $B(4, 0)$, and $C(2, 6)$. (a) Find the length of each side of the triangle. (b) Determine whether the triangle is isosceles, and justify your answer.

(a) $AB = \sqrt{(4-0)^2 + (0-0)^2} = 4$. $AC = \sqrt{(2-0)^2 + (6-0)^2} = \sqrt{4 + 36} = \sqrt{40} = 2\sqrt{10}$. $BC = \sqrt{(2-4)^2 + (6-0)^2} = \sqrt{4 + 36} = \sqrt{40} = 2\sqrt{10}$. (b) Since $AC = BC = 2\sqrt{10}$ (two sides are equal in length), the triangle is isosceles.

14 **[Intermediate]** • **Target: Graphing inequalities in the coordinate plane**

Determine whether the point $(3, 1)$ satisfies the inequality $y < 2x - 3$.

Substitute: $1 < 2(3) - 3 = 3$? True. The point satisfies the inequality.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Graphing inequalities in the coordinate plane • Cross-Lesson: 5.01 Plane Geometry

A region is defined by $y \geq x + 1$. Determine whether $(0, 0)$ lies in this region.

Substitute: $0 \geq 0 + 1 = 1$? False. The point $(0, 0)$ does not lie in the region.

16 [Challenging] • Target: The equation of a parabola in the coordinate plane • Cross-Lesson: 6.01 Trigonometry

Write the equation of a parabola with vertex $(2, -3)$ that passes through $(4, 5)$.

Vertex form: $y = a(x - 2)^2 - 3$. Substitute $(4, 5)$: $5 = a(4 - 2)^2 - 3$, so $5 = 4a - 3$, giving $a = 2$. Equation: $y = 2(x - 2)^2 - 3$.

17 [Challenging] • Target: Finding points that divide a segment in a given ratio • Cross-Lesson: 7.01 Statistics & Probability

Find the point that divides the segment from $(1, 2)$ to $(9, 10)$ in a ratio of $1 : 3$ (closer to the first point).

Using the section formula for a $1 : 3$ ratio: $x = 1 + \frac{1}{4}(9 - 1) = 1 + 2 = 3$. $y = 2 + \frac{1}{4}(10 - 2) = 2 + 2 = 4$. Point: $(3, 4)$.

18 [Challenging] • Target: A multi-part problem combining lines and circles • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

This question has two parts. A circle has equation $(x - 2)^2 + (y + 1)^2 = 25$. (a) State the center and radius. (b) Determine whether the point $(5, 3)$ lies inside, on, or outside the circle, by computing its distance from the center.

(a) Center: $(2, -1)$. Radius: $\sqrt{25} = 5$. (b) Distance from center to $(5, 3)$: $\sqrt{(5 - 2)^2 + (3 - (-1))^2} = \sqrt{9 + 16} = \sqrt{25} = 5$. Since this distance equals the radius, the point $(5, 3)$ lies exactly ON the circle.

19 [Challenging] • Target: Systems of a line and a circle • Cross-Lesson: 1.01 Numbers, Percents & Ratios

Find the points of intersection of the line $y = x + 1$ and the circle $x^2 + y^2 = 25$.

Substitute: $x^2 + (x + 1)^2 = 25$, so $x^2 + x^2 + 2x + 1 = 25$, giving $2x^2 + 2x - 24 = 0$, or $x^2 + x - 12 = 0$. Factor: $(x + 4)(x - 3) = 0$, so $x = -4$ or $x = 3$. Corresponding y-values: for $x = -4$, $y = -3$; for $x = 3$, $y = 4$. Intersection points: $(-4, -3)$ and $(3, 4)$.

20 [Challenging] • Target: Finding area using the coordinate plane • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

Point $M(3, -2)$ is the midpoint of line segment $\overline{AB}$. If point A has coordinates $(-1, 4)$, what are the coordinates of point B?

Using the midpoint formula for each coordinate: $x_M = \frac{x_A + x_B}{2} \Rightarrow 3 = \frac{-1 + x_B}{2} \Rightarrow 6 = -1 + x_B \Rightarrow x_B = 7$. $y_M = \frac{y_A + y_B}{2} \Rightarrow -2 = \frac{4 + y_B}{2} \Rightarrow -4 = 4 + y_B \Rightarrow y_B = -8$. Therefore, the coordinates of B are (7, -8).

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Distance and midpoint formulas

Find the radius of the circle $x^2 + y^2 + 16x - 18y + 81 = 0$, by completing the square.

a. 8 ✓

b. 7

c. 64

d. 9

Completing the square: $(x + 8)^2 + (y - 9)^2 = 64$. Center: $(-8, 9)$. Radius: $\sqrt{64} = 8$.

22 [Foundational] • Target: Slope and equations of lines

Find the equation of the line perpendicular to $y = -\frac{5}{4}x$ passing through $(0, -3)$, in slope-intercept form.

a. $y = -\frac{4}{5}x - 3$

b. $y = \frac{4}{5}x - 3$ ✓

c. $y = -\frac{5}{4}x - 3$

d. $y = \frac{4}{5}x - 2$

The perpendicular slope is the negative reciprocal: $m = \frac{4}{5}$. $y - (-3) = \frac{4}{5}(x - 0)$, giving $y = \frac{4}{5}x - 3$.

23 [Foundational] • Target: Parallel and perpendicular lines

In the standard (x, y) coordinate plane, line l contains the points $(2, 5)$ and $(6, -3)$. Line k is perpendicular to the segment connecting these two points and passes through its midpoint. What is the y-intercept of line k?

a. -6

b. **-1** ✓

c. 3

d. 9

To find the y-intercept of line k, perform the following steps: 1. **Find the midpoint of the segment:** $M = \left(\frac{2+6}{2}, \frac{5+(-3)}{2}\right) = \left(\frac{8}{2}, \frac{2}{2}\right) = (4, 1)$ 2. **Find the slope of line l:** $m_l = \frac{-3-5}{6-2} = \frac{-8}{4} = -2$ 3. **Determine the slope of line k:** Since line k is perpendicular to line l, its slope is the negative reciprocal of m_l: $m_k = -\frac{1}{-2} = \frac{1}{2}$ 4. **Find the y-intercept of line k:** Using the sl.

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24 [Foundational] • Target: The equation of a circle

In the diagram below, lines AB and CD are parallel. Line segment EF is a transversal intersecting AB at G and CD at H. If the measure of angle $\angle AGH$ is 125°, what is the measure of angle $\angle GHD$?

Diagram description: Two parallel horizontal lines AB and CD are intersected by a transversal line EF. Points A and C are on the left, B and D are on the right. G is on AB, H is on CD. Angle AGH is in the upper left quadrant formed by the intersection.

a. 55° ✓

b. 125°

c. 90°

d. 180°

When two parallel lines are intersected by a transversal, consecutive interior angles are supplementary (add up to 180°). Angle $\angle AGH$ and angle $\angle GHC$ are consecutive interior angles. Therefore, $\angle AGH + \angle GHC = 180^\circ$. We are given that $\angle AGH = 125^\circ$. So, $125^\circ + \angle GHC = 180^\circ$. Solving for $\angle GHC$, we get $\angle GHC = 180^\circ - 125^\circ = 55^\circ$. Angle $\angle GHC$ and angle $\angle GHD$ are angles on a straight line (linear pair), so they are.

25 [Foundational] • Target: Reflections, translations, and symmetry

In the coordinate plane, the circle $x^2 + y^2 = 25$ intersects the line $y = 3$ at two points. What is the distance between these two points?

a. 6

b. **8** ✓

c. 10

d. 12

Substitute $y = 3$: $x^2 + 9 = 25 \Rightarrow x^2 = 16 \Rightarrow x = \pm 4$. Points are $(-4, 3)$ and $(4, 3)$. Distance is $4 - (-4) = 8$.

26 [Foundational] • Target: Finding area using the coordinate plane

In ΔABC, the lengths of the sides are $AB = 13$, $BC = 14$, and $AC = 15$. A line segment $\overline{DE}$ is drawn parallel to $\overline{BC}$, with D lying on $\overline{AB}$ and E lying on $\overline{AC}$. If the perimeter of ΔADE is equal to the perimeter of trapezoid $BCED$, what is the length of $\overline{DE}$?

a. 7

b. $\frac{42}{5}$

c. $\frac{21}{2}$ ✓

d. $7\sqrt{2}$

Because $\overline{DE} \parallel \overline{BC}$, $\Delta ADE \sim \Delta ABC$. Let $k = \frac{DE}{BC} = \frac{AD}{AB} = \frac{AE}{AC}$ represent the similarity scale factor, where $0 < k < 1$. 1. Express the side lengths of ΔADE in terms of k: $AD = 13k, \quad AE = 15k, \quad DE = 14k$ The perimeter of ΔADE is: $P_{\Delta ADE} = 13k + 15k + 14k = 42k$ 2. Express the side lengths of trapezoid $BCED$ in terms of k: $DB = AB - AD = 13(1 - k)$ $EC = AC - AE = 15(1 - k)$ $BC = 14$ $DE = 14k$ The.

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27 [Foundational] • Target: Determining if points are collinear

In the coordinate plane, the circle $(x-2)^2+(y-3)^2=25$ intersects the y-axis at two points. What is the distance between these two points?

a. $2\sqrt{21}$ ✓

b. $4\sqrt{5}$

c. 8

d. 10

For the y-axis, set $x=0$: $(0-2)^2+(y-3)^2=25 \Rightarrow 4+(y-3)^2=25 \Rightarrow (y-3)^2=21 \Rightarrow y=3\pm\sqrt{21}$. The distance between the points is $(3+\sqrt{21})-(3-\sqrt{21})=2\sqrt{21}$.

28 [Foundational] • Target: Finding the intersection of two lines

Two parallel lines are cut by a transversal. One angle measures 105°. What is its alternate exterior angle?

a. 168°

b. 67°

c. 91°

d. **105°** ✓

Alternate exterior angles are congruent: 105°.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Combining coordinate geometry concepts

Two parallel lines are cut by a transversal. One angle measures 86°. What is its alternate interior angle?

a. 124°

b. **86°** ✓

c. 119°

d. 176°

Alternate interior angles are congruent: 86°.

30 [Intermediate] • Target: Graphing inequalities in the coordinate plane

Two parallel lines are cut by a transversal. One angle measures 130°. What is its co-interior (same-side interior) angle?

a. 12°

b. 108°

c. 100°

d. **50°** ✓

Co-interior angles are supplementary: $180-130=50$.

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31 [Intermediate] • Target: The equation of a parabola in the coordinate plane

Two parallel lines are cut by a transversal. One angle measures 60°. What is its corresponding angle?

a. **60°** ✓

b. 74°

c. 144°

d. 118°

Corresponding angles are congruent: 60°.

32 [Intermediate] • Target: Finding points that divide a segment in a given ratio

Two parallel lines are cut by a transversal. One angle measures 120°. What is its vertical angle?

a. 162°

b. **120°** ✓

c. 139°

d. 11°

Vertical angles are congruent: 120°.

33 [Intermediate] • Target: A multi-part problem combining lines and circles

In the standard (x, y) coordinate plane, the vertices of ΔOAB are $O(0, 0)$, $A(6, 0)$, and $B(2, 4)$. A vertical line with equation $x = k$, where $0 < k < 6$, divides ΔOAB into two regions of equal area. What is the value of k?

a. $6 - 2\sqrt{3}$ ✓

b. $\sqrt{6}$

c. $\frac{8}{3}$

d. 3

1. **Find the total area of ΔOAB:** Taking the side along the x-axis as the base, the length of base OA is $6 - 0 = 6$. The height is the y-coordinate of vertex $B(2, 4)$, which is 4. $\text{Area}(\Delta OAB) = \frac{1}{2} \times 6 \times 4 = 12$ To divide the triangle into two regions of equal area, each region must have an area of $\frac{12}{2} = 6$. 2. **Determine the domain of k:** - If 02, the line $x = k$ lies to the right of vertex $B(2, 4)$. The region to the right of $x = k$ is a t.

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34 [Intermediate] • Target: Systems of a line and a circle

In the standard (x, y) coordinate plane, the line $y = 2x - 3$ intersects the parabola $y = x^2 - 6x + c$ at two distinct points, P and Q. If the distance between P and Q is $4\sqrt{5}$, what is the value of the constant c?

a. -7

b. 9 ✓

c. 12

d. 15

To find the points of intersection, set the two equations equal: $x^2 - 6x + c = 2x - 3$ Rearranging into standard quadratic form gives: $x^2 - 8x + (c + 3) = 0$ Let the roots of this equation be x_1 and x_2. By Vieta's formulas, the sum and product of the roots are: $x_1 + x_2 = 8$ and $x_1x_2 = c + 3$ The horizontal distance between the intersection points is given by $|x_2 - x_1|$. We can calculate $(x_2 - x_1)^2$ using the identity.

35 [Intermediate] • Target: Distance and midpoint formulas

The graph of the quadratic function $f(x) = x^2 - 6x + 5$ is shown in the standard (x, y) coordinate plane below. A horizontal line $y = k$ intersects the graph of $f(x)$ at two points that are 8 units apart. What is the value of k?

a. 12 ✓

b. 8

c. 4

d. -3

We are given the quadratic function $f(x) = x^2 - 6x + 5$ and a horizontal line $y = k$ that intersects the parabola at two points separated by a horizontal distance of 8. **Method 1: Using the axis of symmetry** The axis of symmetry for a parabola in standard form $y = ax^2 + bx + c$ is given by: $x = -\frac{b}{2a} = -\frac{-6}{2(1)} = 3$ Because parabolas are symmetric about their axis of symmetry, the two intersection points on the horizontal line $y = k$ must lie equidistant from $x = 3$. Since the total dist.

36 [Intermediate] • Target: Slope and equations of lines

The line $x = 5$ intersects the circle $(x - 2)^2 + (y + 1)^2 = 25$ at two points. What is the distance between these two points?

a. 6

b. **8** ✓

c. 10

d. 12

Substitute $x = 5$: $(5-2)^2 + (y+1)^2 = 25 \Rightarrow 9 + (y+1)^2 = 25 \Rightarrow (y+1)^2 = 16 \Rightarrow y + 1 = \pm 4 \Rightarrow y = 3$ or $y = -5$. Distance $= 3 - (-5) = 8$.

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37 [Intermediate] • Target: Parallel and perpendicular lines

A line has slope -2. What is the slope of a line perpendicular to it?

a. $\frac{1}{10}$

b. $\frac{2}{9}$

c. $\frac{1}{2}$ ✓

d. $\frac{3}{4}$

Perpendicular slopes are negative reciprocals: $-1/(-2) = \frac{1}{2}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The equation of a circle • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

The parabola $y = x^2 - x - 2$ crosses the x-axis at two points. What is the sum of their x-coordinates?

a. -4

b. **1** ✓

c. -2

d. 5

By Vieta's formulas, the sum of the roots is $-b/a = 0 + 1 = 1$.

39 [Challenging] • Target: Reflections, translations, and symmetry • Cross-Lesson: 1.01 Numbers, Percents & Ratios

A line has slope 2. What is the slope of a line parallel to it?

a. $\frac{51}{100}$

b. 2 ✓

c. $\frac{13}{20}$

d. $\frac{1}{2}$

Parallel lines have equal slope: 2.

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40 [Challenging] • Target: Finding area using the coordinate plane • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

In the standard (x, y) coordinate plane, a triangle has vertices at $A(0, 0)$, $B(8, 0)$, and $C(2, 6)$. A vertical line with equation $x = k$, where $0 < k < 8$, divides the triangle into two regions of equal area. What is the value of k?

a. $8 - 2\sqrt{6}$ ✓

b. 4

c. $\frac{10}{3}$

d. $2\sqrt{6}$

1. **Find the total area of ΔABC:** The base along the x-axis connects $A(0, 0)$ and $B(8, 0)$, so the base length is $b = 8$. The vertex $C(2, 6)$ gives a height of $h = 6$. Area$(\Delta ABC) = \frac{1}{2} \times 8 \times 6 = 24$ Each of the two regions must therefore have an area of $\frac{24}{2} = 12$. 2. **Determine the location of $x = k$:** The vertical line through C is $x = 2$. The portion of ΔABC to the left of $x = 2$ is a triangle with base on the x-axis from $x = 0$ to.

41 [Challenging] • Target: Determining if points are collinear • Cross-Lesson: 3.01 Functions & Graphs

In the xy-plane, the points $(1, 3)$ and $(7, 11)$ are the endpoints of a diameter of a circle. What is the equation of the tangent line to the circle at the point $(7, 11)$?

a. $3x + 4y = 65$ ✓

b. $4x + 3y = 61$

c. $3x - 4y = -23$

d. $4x - 3y = -5$

Center is $\left(\frac{1+7}{2}, \frac{3+11}{2}\right) = (4, 7)$. Slope of radial segment from center to $(7, 11)$ is $\frac{11-7}{7-4} = \frac{4}{3}$. The tangent line is perpendicular to the radius, so its slope is $-\frac{3}{4}$. Using point $(7, 11)$:

$y - 11 = -\frac{3}{4}(x - 7) \Rightarrow 4y - 44 = -3x + 21 \Rightarrow 3x + 4y = 65$.

42 [Challenging] • Target: Finding the intersection of two lines • Cross-Lesson: 5.01 Plane Geometry

The line $y = 3x - 5$ intersects the parabola $y = x^2 - 2x + 1$ at points P and Q. What is the x-coordinate of the midpoint of segment PQ?

a. 2

b. 2.5 ✓

c. 3

d. 3.5

Set equal: $x^2 - 2x + 1 = 3x - 5 \Rightarrow x^2 - 5x + 6 = 0$. By Vieta's formulas, the sum of the x-coordinates is 5. The x-coordinate of the midpoint is the average: $\frac{5}{2} = 2.5$.

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43 [Challenging] • Target: Combining coordinate geometry concepts • Cross-Lesson: 6.01 Trigonometry

In the standard (x, y) coordinate plane, the vertices of ΔABC are $A(0, 0)$, $B(10, 0)$, and $C(2, k)$, where $k > 0$. If $\angle ACB = 90^\circ$, what is the value of k?

a. $2\sqrt{2}$

b. **4** ✓

c. $2\sqrt{5}$

d. 8

We are given the vertices $A(0, 0)$, $B(10, 0)$, and $C(2, k)$ with $k > 0$, and that $\angle ACB = 90^\circ$. **Method 1: Slopes of Perpendicular Lines** Since $\angle ACB = 90^\circ$, segment AC is perpendicular to segment BC. The product of their slopes must be -1: - Slope of $AC = \frac{k-0}{2-0} = \frac{k}{2}$ - Slope of $BC = \frac{k-0}{2-10} = -\frac{k}{8}$ Setting the product of their slopes equal to -1: $\left(\frac{k}{2}\right)\left(-\frac{k}{8}\right) = -1$.

44 [Challenging] • Target: Graphing inequalities in the coordinate plane • Cross-Lesson: 7.01 Statistics & Probability

In the standard (x, y) coordinate plane, the complex number $z = x + yi$ satisfies the equation $z(2 - i) = 15 + 5i$ where $i^2 = -1$, and x and y are real numbers. Vector $\mathbf{v}$ is defined by $\mathbf{v} = [x; y]$. If matrix $A = [3, -1; 2, 4]$, what is the magnitude of the vector $A\mathbf{v}$?

a. $5\sqrt{2}$

b. $10\sqrt{5}$

c. $10\sqrt{10}$ ✓

d. 40

Step 1: Solve for the complex number $z = x + yi$. Multiply the numerator and denominator by the complex conjugate of $2 - i$, which is $2 + i$: $z = \frac{15+5i}{2-i} = \frac{(15+5i)(2+i)}{(2-i)(2+i)} = \frac{30+15i+10i+5i^2}{4-i^2}$ Since $i^2 = -1$: $z = \frac{30+25i-5}{4-(-1)} = \frac{25+25i}{5} = 5 + 5i$ Thus, $x = 5$ and $y = 5$. Step 2: Construct vector $\mathbf{v}$ and perform the matrix-vector multiplication $A\mathbf{v}$: $\mathbf{v} = [5; 5]$.

45 [Challenging] • Target: The equation of a parabola in the coordinate plane • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

In the xy-plane, a line passes through the center of the circle $(x - 3)^2 + (y + 2)^2 = 16$ and is perpendicular to $2x - 4y = 9$. What is the y-intercept of this line?

a. $(0, 4)$ ✓

b. $(0, 2)$

c. $(0, -2)$

d. $(0, -8)$

Center of circle is $(3, -2)$. Slope of given line: $4y = 2x - 9 \Rightarrow m = 1/2$. The perpendicular line has slope $m_\perp = -2$. Equation: $y - (-2) = -2(x - 3) \Rightarrow y + 2 = -2x + 6 \Rightarrow y = -2x + 4$. The y-intercept is $(0, 4)$.

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